

21ECO109T/ 5G Technology - An Overview

CASE STUDY PROJECT REPORT

Submitted by

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RAMAPURAM, CHENNAI

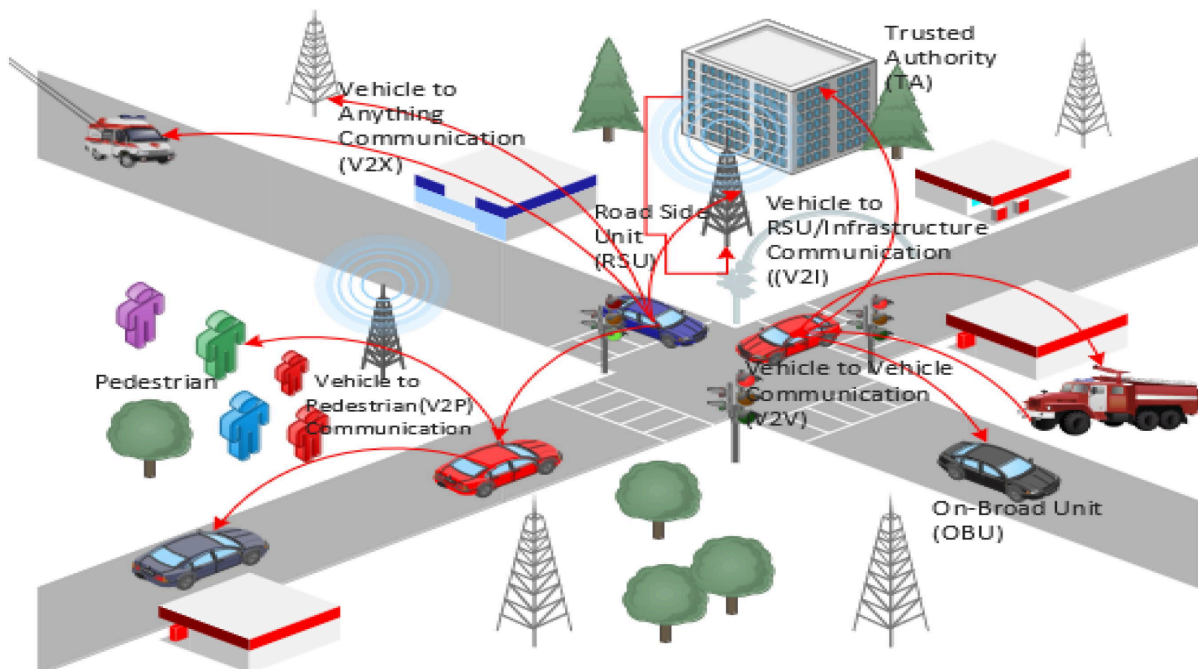
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TOPIC 1: Wireless Communication in Autonomous Vehicles

Introduction

Autonomous vehicles (AVs), also known as self-driving cars, are vehicles capable of sensing their environment and operating without human intervention. Wireless communication plays a vital role in enabling these vehicles to exchange information with other vehicles, road infrastructure, pedestrians, and cloud servers. This communication improves safety, traffic efficiency, and driving comfort.

Modern autonomous vehicles use technologies such as 5G, Wi-Fi, GPS, radar, LiDAR, cameras, and artificial intelligence to make driving decisions in real time.



Components of an Autonomous Vehicle Communication System

Major Components

- Cameras
- LiDAR
- Radar
- Ultrasonic Sensors
- GPS
- Inertial Measurement Unit (IMU)
- Vehicle Control Unit
- Wireless Communication Module (5G/Wi-Fi)

Working

1. Sensors collect environmental information.
2. AI processes sensor data.
3. Wireless communication exchanges information.
4. Decision-making module plans movement.
5. The vehicle performs steering, braking, and acceleration.

Types of Vehicle Communication

Vehicle-to-Vehicle (V2V)

Vehicles share

- Speed
- Position
- Direction
- Brake status

Benefits

- Collision avoidance
- Blind spot warning
- Cooperative driving

Vehicle-to-Infrastructure (V2I)

Communication between vehicles and traffic infrastructure.

Examples

- Traffic lights
- Road signs
- Parking systems

Advantages

- Reduced congestion
- Smart traffic management

Vehicle-to-Pedestrian (V2P)

Communicates with smartphones and wearable devices.

Purpose

- Warn drivers
- Protect pedestrians

Vehicle-to-Network (V2N)

Connects vehicles to cloud services.

Applications

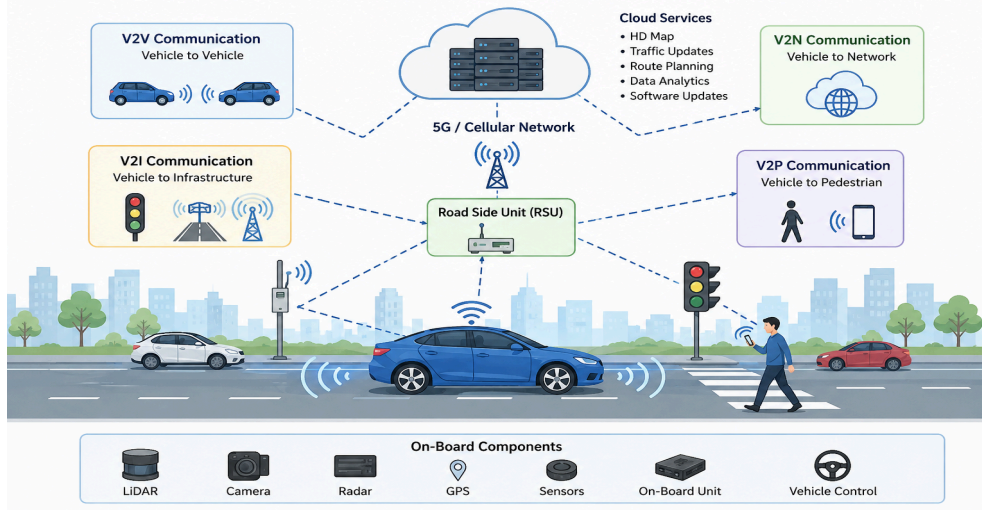
- Navigation
- Weather
- Software updates
- Entertainment

TECHNOLOGY	PURPOSE
5G	High-speed communication
Wi-Fi	Local Connectivity
GPS	Location Tracking
DSRC	Vehicle Communication
Bluetooth	Short Range Wireless communication

Applications

- Self-driving taxis
- Smart parking
- Fleet management
- Emergency vehicle priority
- Highway automation

WIRELESS COMMUNICATION IN AUTONOMOUS VEHICLES



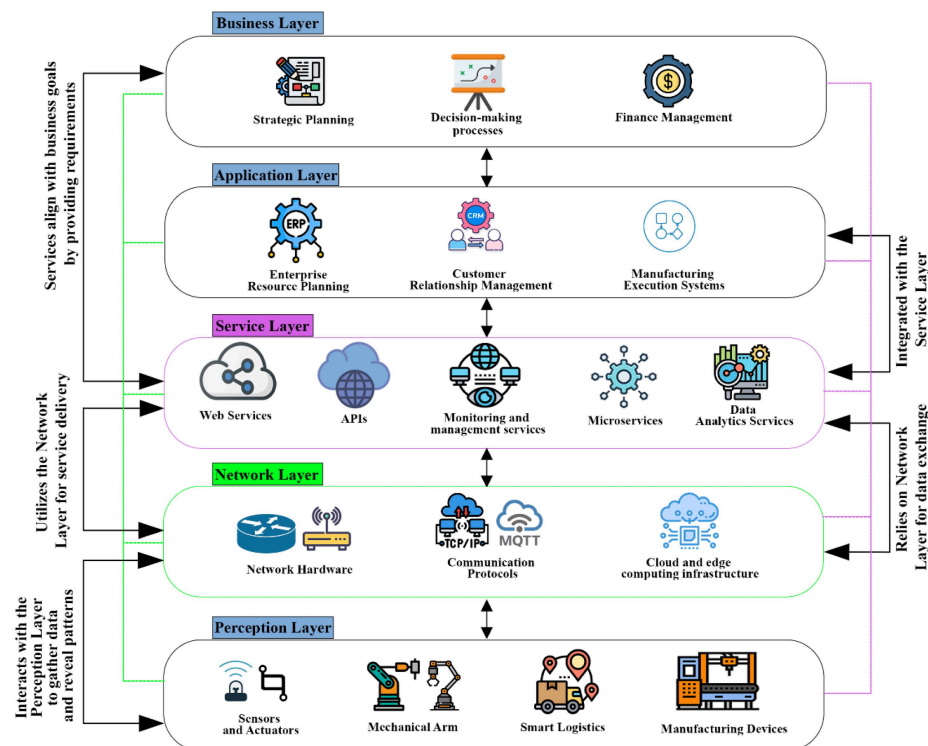
TOPIC 2: 5G Enabled Industrial Automation

5G Enabled Industrial Automation

Introduction

Industrial automation uses intelligent machines, sensors, robots, and communication systems to perform manufacturing tasks with minimal human intervention. The introduction of 5G technology has transformed Industry 4.0 by enabling ultra-fast, reliable, and low-latency communication between machines.

5G supports real-time monitoring, predictive maintenance, remote control, and intelligent decision-making, making factories more productive and efficient.



Working Principle

The industrial automation process using 5G follows these steps:

1. Smart sensors continuously collect information such as temperature, pressure, vibration, and machine status.
2. The collected data is transmitted through the 5G network with ultra-low latency.
3. Edge computing devices process critical data instantly.
4. AI algorithms analyze the data to identify faults or optimize production.
5. Control commands are sent back to robots and machines in real time.
6. Machines automatically adjust their operations without human intervention.

This entire communication occurs within milliseconds, ensuring smooth and uninterrupted industrial operations.

Architecture of a 5G Smart Factory

Components

- Industrial Robots
- IoT Sensors
- 5G Base Station
- Edge Computing
- Cloud Platform

Benefits

- Low latency
- High reliability
- Increased productivity
- Reduced downtime
- Better safety
- Energy efficiency

Applications

- Smart factories
- Warehouse automation
- Predictive maintenance
- Remote machine monitoring
- Automated quality inspection
- Digital twins

5G	Fast Communication
IoT	Data Collection
AI	Decision Making
Edge Computing	Local preprocessing
Robotic	Automation

